

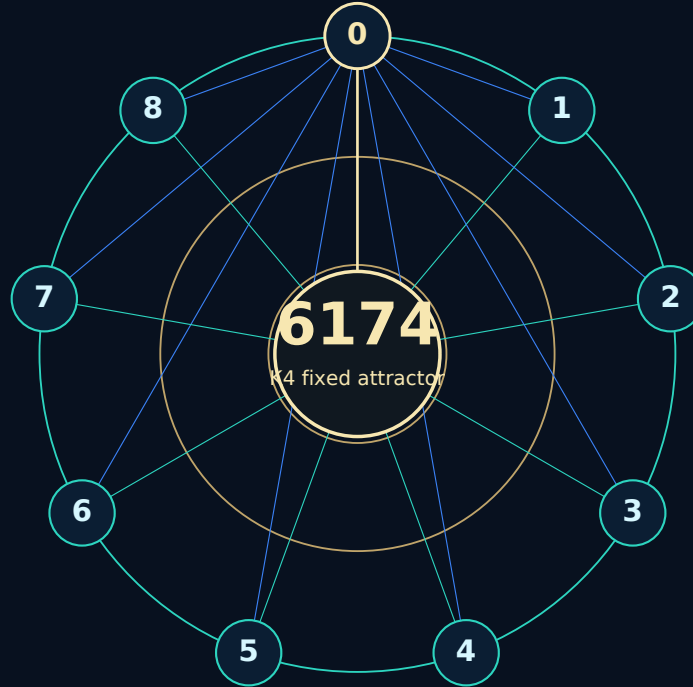
K9 Attractor Theory

Invariant-Constrained Attractor Dynamics for Bounded Decision Systems

Updated with the K2-K12 Kaprekar toy-model ladder, local cycle verification, and expanded attractor ecology.

K9 Attractor Geometry

Residues collapse into the 0 mod 9 channel; iteration exposes the attractor.



$$K(n) = \text{desc}(n) - \text{asc}(n) \equiv 0 \pmod{9}$$

K4 attractor = 6174

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# CANON_HEADER v1.0.0
# NAME=K9_ATTRACTOR_THEORY_INVARIANT_CONSTRAINED_ATTRACTOR_DYNAMICS
# TITLE=K9 Attractor Theory - Invariant-Constrained Attractor Dynamics
# VERSION=0.2.0
# STATUS=DRAFT_RESEARCH_FRAMING
# AUTHORSHIP=TOMS, JANSONS
# AUTHOR_METADATA=TOMS, JANSONS
# ROLE=THEORY_FRAMING_AND_RESEARCH_PROGRAM
# PURPOSE=Preserve the v0.1 base while embedding K2-K12 Kaprekar toy-model attractors and updating the formal/general ladder.
# SUPERSEDES=K9_Attractor_Theory_TOMS_JANSONS_v0_1.pdf
# CHANGE_TYPE=MINOR_EXPANSION
# CHANGE_SUMMARY=Adds K2 cycle, K3 495, K4 6174, K5+ attractor ecology, local computation receipt, and updated requirements/coherence r
# SETS_BOUND=KAPREKAR_BASE;MOD9_RESIDUE;K2_CYCLE;K3_495;K4_6174;K5_PLUS_ATTRACTOR_ECOLOGY
# SETS_BOUND_CONT=K9_ATTRACTOR_FORM;ICAD;GOLDLEVEL;LEADLEVEL;CSM144G;TEMPORAL_GOVERNANCE;5D_CONSTRAINT_MEMORY;NOBEL_CALIBER_RESEARCH_TR
# CLAIM_STATUS=CONCEPTUAL_SYNTHESIS_WITH_LOCAL_TOY_MODEL_COMPUTATION_NOT_EXTERNAL_VALIDATION
# OUTPUT_FORMAT=PDF
# FILE_METADATA_AUTHOR=TOMS, JANSONS
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# CANONICAL=false
```

PDF metadata author is set to TOMS, JANSONS. The same authorship is present in the canonical header, footer, and change-control record.

0. Document Control and Requirements Register

This v0.2 update preserves the v0.1 core base and embeds the new K2/K3/K4/K5+ Kaprekar model material as additional sections, updated tables, and expanded model logic. It does not replace the original K9 Attractor Theory frame; it extends the toy-model foundation beneath it.

Boundary: this PDF frames a research direction and documents locally computed toy-model cycles. It does not claim peer review, Nobel assessment, external validation, or real-world economic proof.

ID	Requirement	Fulfillment location	Status
R1	Update the previous PDF while maintaining the core base.	v0.2 canonical header, change control, Sections 1-10 retain v0.1 structure and add new ladder material.	Fulfilled
R2	Embed the K2 attractor, K3 model, K4/6174 model, and K5+ models.	Sections 2A, 2B, 4, and Appendix A.	Fulfilled
R3	Outline the models in table form.	Main summary table in Section 2A; detailed cycle inventory in Appendix A.	Fulfilled
R4	Preserve authorship TOMS, JANSONS in metadata and header.	PDF metadata, cover header, footer, and change control.	Fulfilled
R5	Keep requirements documented throughout the file.	Requirement register, source-status split, local verification receipt, claim boundary register, coherence record.	Fulfilled
R6	Update previous sections where the new material changes the interpretation.	Formal model, geometric mapping, validation program, applications, and roadmap updated for K2-K12 attractor ecology.	Fulfilled
R7	Avoid overclaiming.	Non-claim register, evidence gates, and claim labels maintained and expanded.	Fulfilled

Static Table of Contents

Section	Title	Function
0	Document Control and Requirements Register	Shows how the update request is fulfilled.
1	Source Status and Evidence Discipline	Maintains the known/inferred/not-proven split.
2	Base Lesson	Restates Kaprekar/mod 9 as the primitive.
2A	Adjacent Toy-Model Ladder	Adds K2 through K12 attractor summary and computation receipt.
2B	Detailed K2/K3/K4/K5+ Interpretation	Explains the structural break from clean constants to cycle ecology.
3	New Form	Defines K9 Attractor Theory as possibility discovery under pressure.
4	Formal Model	Updates ICAD with K_n and attractor-class taxonomy.
5	Geometric Pattern	Extends geometry from single center to cycle/ecology views.
6	Nobel-Caliber Framing	Frames the research track without claiming prize association.
7	Validation Program	Defines what would be needed for serious external recognition.
8	Governance and Claim Safety	Keeps proof/state boundaries intact.
9	Applications	Shows business, AI, trust, institutional, and personal templates.
10	Roadmap, Change Control, Coherence	Documents version and QA record.
A	Appendix A: K2-K12 Cycle Inventory	Full local toy-model cycle table.

1. Source Status and Evidence Discipline

The updated artifact combines three source layers: the mathematical Kaprekar routine, the conversation-developed K9/ICAD conceptual frame, and internal governance discipline for evidence, versioning, state, and claim safety.

1.1 Source classes

Source class	Used for	Status in this document
Mathematical primitive	Kaprekar sorting/subtraction routine; fixed-length digit strings; leading zeros retained.	Elementary arithmetic model.
Local computation	Enumeration of sorted digit-multiset states for n=2..12 to identify cycles.	Internal computation receipt, not peer review.
Thread synthesis	K9 Attractor Form; mod 9 energy-flow metaphor; Nobel-worthy framing language.	Conceptual synthesis.
CSM144G kernel	Evidence discipline, proof boundary, state promotion, artifact authority.	Internal project authority, not public validation.
GOLDLEVEL versioning	v0.1 -> v0.2 as a minor expansion preserving the base.	Internal version discipline.
Nobel-caliber framing	Economic/decision-systems research ambition.	Hypothesis and positioning only.

1.2 Evidence gates

The update follows the rule that model output can structure the artifact, but proof, state promotion, receipts, and external recognition remain outside the prose artifact. The K2-K12 cycles are locally computed and documented as a toy-model receipt; the general ICAD theory remains draft research framing.

1.3 Non-claim register

Statement not made	Reason
This work has won or is associated with a Nobel Prize.	No such claim is evidenced.
K9 Attractor Theory is established science.	It remains a draft research program.
K2-K12 local cycles prove the full decision-system theory.	They only verify finite arithmetic toy models.
Mod 9 energy-flow is literal physical energy.	It is metaphorical dynamics language unless formalized otherwise.
The model predicts markets today.	No empirical benchmark study has been performed.

2. Base Lesson: Kaprekar as the Toy Model

Kaprekar's four-digit routine is the primitive pattern: sort digits descending, sort digits ascending, subtract, and iterate. For non-repeating four-digit inputs, the process enters 6174. The mod 9 rule appears because digit rearrangement preserves digit sum, so the subtraction of two rearrangements yields a multiple of 9.

$$K_n(x) = desc_n(x) - asc_n(x)$$
$$desc_n(x) \equiv asc_n(x) \pmod{9}$$

Therefore: $K_n(x) \equiv 0 \pmod{9}$ after one nontrivial operation

For K4 with sorted digits $a \geq b \geq c \geq d$:

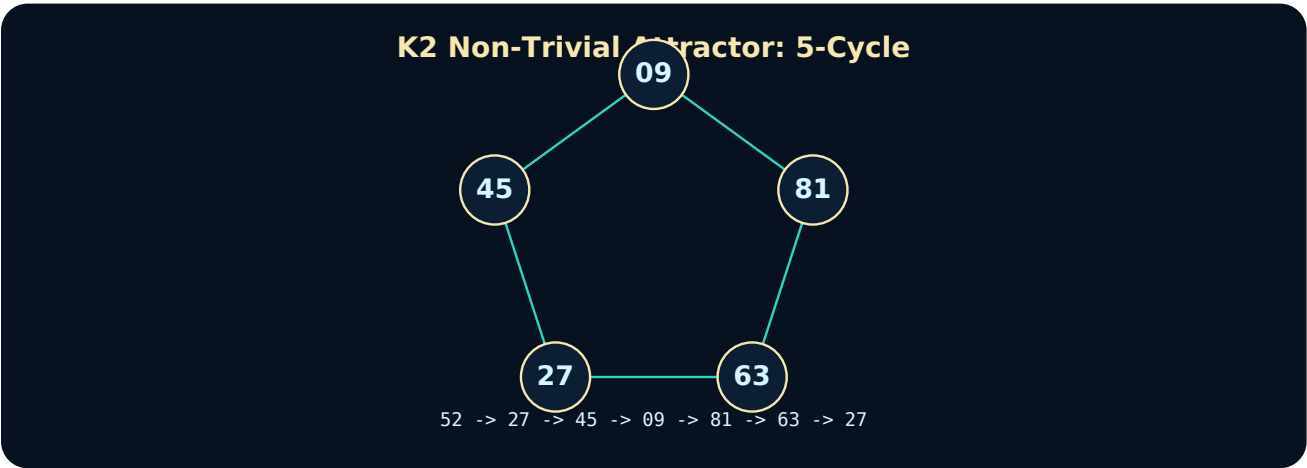
$$K_4(n) = (1000a + 100b + 10c + d) - (1000d + 100c + 10b + a)$$
$$= 999(a-d) + 90(b-c)$$
$$= 9[111(a-d) + 10(b-c)]$$

The base lesson is now enlarged: K4/6174 is the cleanest famous case, but the adjacent models show that finite symbolic systems may converge to a fixed point, a cycle, or an ecology of multiple attractors.

Kaprekar element	Generalized lesson
Digit length n	Bounded symbolic system size.
Descending sort	Dominant order or visible strongest-to-weakest structure.
Ascending sort	Inverted or shadow order.
Subtraction	Operational difference between expression and inversion.
Mod 9 multiple	Invariant or forced channel after operation.
Iteration	Time, feedback, pressure, repeated observation.
Fixed point/cycle/ecology	Attractor class discovered by the system.

2A. Adjacent Toy-Model Ladder: K2 through K12

The closest adjacent toy model to K4/6174 is K3/495. The next lower model, K2, does not have a single non-trivial fixed constant; it enters a 5-cycle. K5 and above break from the clean single-constant pattern into multiple attractor basins and cycles.



Model	Digit length	Non-trivial attractor behavior	Representative attractor/cycle
K2	2	One non-trivial 5-cycle; repdigits collapse to zero.	09 -> 81 -> 63 -> 27 -> ...
K3	3	Single fixed attractor.	495
K4	4	Single fixed attractor.	6174
K5	5	First multi-cycle ecology: one 2-cycle plus two 4-cycles.	53955 -> 59994; 61974 -> 82962 -> 75933 -> 63954
K6	6	Attractor ecology: 3 non-trivial cycle(s), lengths [1, 1, 7].	549945; 631764
K7	7	Attractor ecology: 1 non-trivial cycle(s), lengths [8].	7509843 -> 9529641 -> 8719722 -> 8649432 -> ...
K8	8	Attractor ecology: 4 non-trivial cycle(s), lengths [1, 1, 3, 7].	63317664; 97508421
K9	9	Attractor ecology: 3 non-trivial cycle(s), lengths [1, 1, 14].	554999445; 864197532
K10	10	Attractor ecology: 8 non-trivial cycle(s), lengths [1, 1, 1, 3, 3, 3, 3, 7].	6333176664; 9753086421
K11	11	Attractor ecology: 3 non-trivial cycle(s), lengths [1, 5, 8].	86431976532; 86420987532 -> 96641975331 -> 88431976512 -> 87641975322 -> ...
K12	12	Attractor ecology: 16 non-trivial cycle(s), lengths [1, 1, 1, 1, 1, 3, 3, 3, 3, 3, 3, 3, 3, 3, 7].	555499994445; 633331766664

Local computation receipt: cycles were generated by enumerating sorted digit multisets of length n, applying $K_n(x)=desc_n(x)-asc_n(x)$, padding with leading zeros, re-sorting, and detecting cycles in the finite state graph. Full inventory appears in Appendix A.

2B. Detailed K2/K3/K4/K5+ Interpretation

2B.1 K2: first non-trivial cycle model

K2 shows that the smallest non-trivial Kaprekar system does not converge to a single constant. It collapses either to 00 for repdigits or into the cycle $09 \rightarrow 81 \rightarrow 63 \rightarrow 27 \rightarrow 45 \rightarrow 09$. This is the first warning against assuming every finite symbolic system has one center.

Example: $52 - 25 = 27$
 $72 - 27 = 45$
 $54 - 45 = 09$
 $90 - 09 = 81$
 $81 - 18 = 63$
 $63 - 36 = 27$

2B.2 K3: minimal fixed-attractor model

K3 is the clean minimal constant-attractor model. For valid three-digit inputs, the routine reaches 495.

$753 - 357 = 396$
 $963 - 369 = 594$
 $954 - 459 = 495$
 $954 - 459 = 495$

2B.3 K4: canonical fixed-attractor model

K4 gives the famous fixed attractor 6174. It is pedagogically powerful because its basin is unusually clean for four digits, while still demonstrating sorting, inversion, invariant collapse, and convergence.

$7174 \rightarrow 7741 - 1477 = 6264$
 $6264 \rightarrow 6642 - 2466 = 4176$
 $4176 \rightarrow 7641 - 1467 = 6174$

2B.4 K5+: attractor ecology

K5 and higher models show the more general pattern: not one center, but a basin ecology. Systems can have fixed points, 2-cycles, 3-cycles, 5-cycles, 7-cycles, 8-cycles, 14-cycles, and multiple competing basins. This is the conceptual bridge from recreational arithmetic to decision-system theory.

3. New Form: The K9 Possibility Attractor

K9 Attractor Theory generalizes the toy models into a method for discovering what a bounded system becomes under pressure. The new K2-K12 ladder strengthens the base: the result is not always a single point. The possible output classes are fixed points, cycles, oscillations, basin ecologies, degradation, and terminal collapse.

3.1 Core thesis

In finite systems where signals can be ranked, inverted, compared, and iterated, the system's true structure is not revealed by its current state alone. It is revealed by its attractor: the fixed, cyclic, degrading, terminal, or multi-basin form it repeatedly becomes under pressure.

3.2 General sequence

Input chaos -> ordered polarity -> inversion
-> subtractive difference -> invariant channel
-> iteration through time -> attractor class

3.3 Possibility statement

Possibility is the reachable state-space remaining after repeated sorting, inversion, differencing, constraint memory, and feedback. K2 teaches cycle. K3 and K4 teach fixed convergence. K5+ teaches basin ecology. Together, they define the broader possibility space.

Attractor class	Mathematical toy example	General decision-system meaning
Zero collapse	Repdigits -> 00...0	No usable differentiation or insufficient signal.
Fixed point	K3=495; K4=6174	Stable repeating identity under pressure.
Short cycle	K2 5-cycle; K5 2-cycle	Pattern repeats without full stabilization.
Long cycle	K7 8-cycle; K9 14-cycle	Complex repeating attractor after many transitions.
Multi-basin ecology	K5, K8, K10, K12	Several possible futures depending on entry state.
Terminal/degrading state	Not arithmetic; governance overlay	System cannot continue safely without repair or termination.

4. Formal Model: Invariant-Constrained Attractor Dynamics

ICAD now explicitly distinguishes between the Kaprekar toy family K_n and the generalized decision-system model. The toy family supplies finite-state examples; the general model supplies the research hypothesis.

4.1 Kaprekar toy-family operator

Let x be an n -digit base-10 string, leading zeros retained.
 $K_n(x) = \text{desc}_n(x) - \text{asc}_n(x)$, padded back to n digits.
State for graph computation = sorted digits of $K_n(x)$.
Attractor = fixed point or directed cycle reached by repeated K_n .

4.2 General ICAD operator

Let $S_t = [s_1, s_2, \dots, s_m]$ be a bounded signal vector.
 $\text{Rank}(S_t)$ = strongest-to-weakest ordering.
 $\text{Invert}(S_t)$ = weakest-to-strongest ordering.
 $D_t = \text{Difference}(\text{Rank}(S_t), \text{Invert}(S_t))$.
 I_t = invariant or conserved bottleneck detected across iterations.
 A = attractor class reached or repeatedly approached by S_t under pressure.

4.3 Attractor-class taxonomy

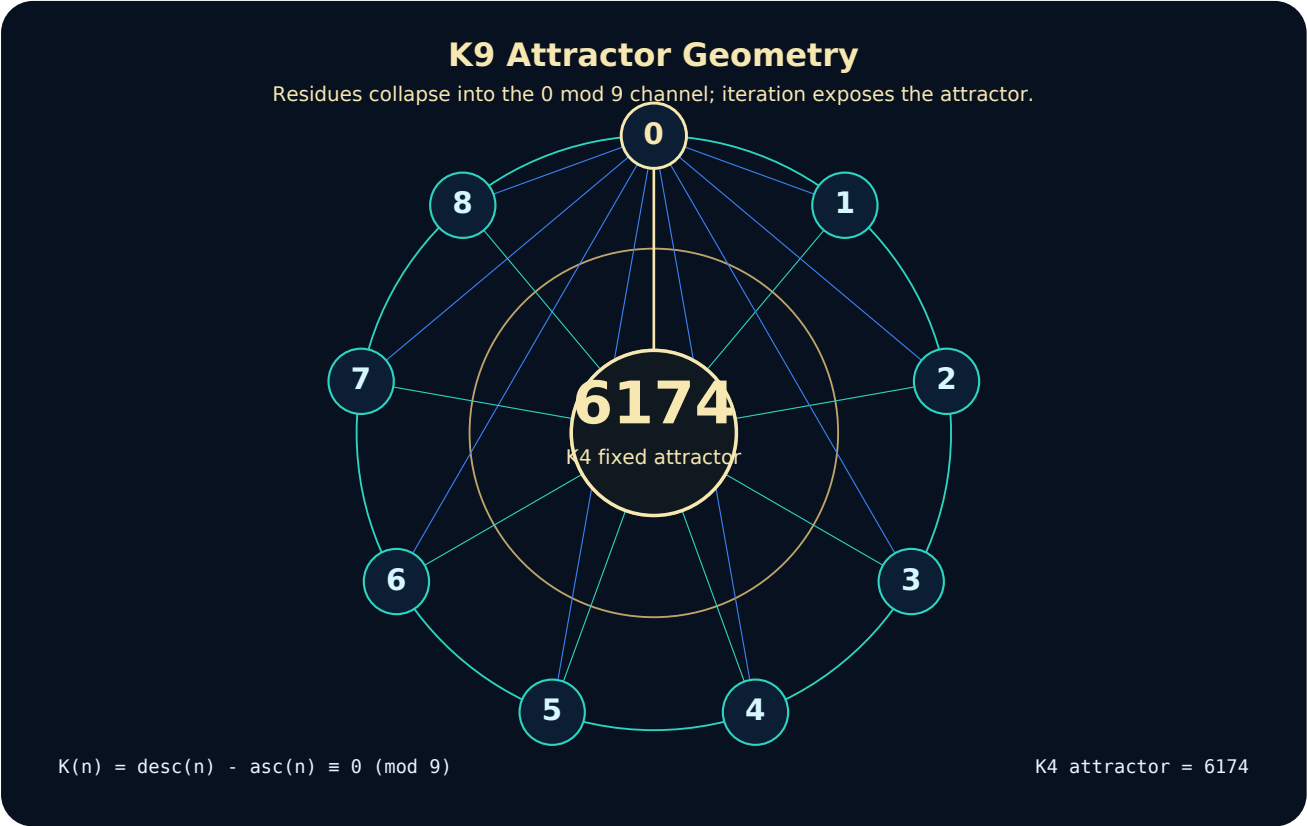
Class	Toy signature	Decision-system signature
Zero collapse	00...0	No signal differentiation; all forces equal or inert.
Fixed point	495, 6174	Stable repeatable operating form.
Cycle	K2 5-cycle	Repeated unresolved loop.
Ecology	K5+ multiple basins	Different outcomes from different starting states.
Degradation	Governance overlay	Trajectory worsens under iteration.
Terminal	Governance overlay	Further action causes waste, harm, or invalidity.

4.4 Why K5+ matters

K4 is memorable because it is clean. K5+ is important because it is realistic. Real decision systems rarely have one universal attractor; they have several reachable basins shaped by entry conditions, constraints, feedback, and history.

5. Geometric Pattern: From Single Center to Attractor Ecology

The v0.1 visual used a central 6174 attractor inside a mod 9 field. The v0.2 update preserves that geometry while adding adjacent forms: cycle geometry for K2, fixed-center geometry for K3/K4, and basin ecology for K5+.



Model	Geometry	Interpretive use
K2	Pentagonal directed cycle	Teaches loop rather than center.
K3	Single fixed center: 495	Teaches minimal fixed convergence.
K4	Single fixed center: 6174	Teaches canonical attractor convergence.
K5+	Multiple centers and cycles	Teaches basin ecology and path dependence.
ICAD	State-space map with constraints	Teaches decision trajectories under pressure.

Geometry is not decoration here. It is a compression method: a diagram that forces the operator to ask what the strongest is, where the system flows, and what repeated form appears.

6. Nobel-Caliber Framing

The strongest framing remains Economic Sciences or decision science rather than pure mathematics. The K2-K12 ladder strengthens the research frame by showing that the seed is not a single curiosity but a family of finite symbolic attractor systems.

6.1 Formal-title candidates

Use case	Title
Academic	Invariant-Constrained Attractor Dynamics for Bounded Decision Systems
Economic science	Attractor Diagnostics for Coordination Failure, Trust Decay, and Resource Misallocation
AI governance	Constraint-Aware Convergence Detection for Human-AI Decision Systems
Toy-model bridge	From Kaprekar Cycles to Multi-Basin Decision Attractors

6.2 Nobel-worthy abstract

This work introduces Invariant-Constrained Attractor Dynamics, a framework for modeling finite decision systems under repeated pressure. Inspired by the Kaprekar family of digit-sorting routines, the framework shows that ranking, inversion, differencing, and iteration expose fixed points, cycles, and attractor ecologies. These classes provide a possible way to detect convergence, oscillation, degradation, and terminal collapse in economic coordination, institutional trust, AI-mediated workflows, and bounded strategic systems before conventional metrics register failure.

6.3 Prize-track logic

Layer	Required contribution	Prize-caliber condition
Mathematical	Define K_n toy models, ICAD operators, invariants, basins, and convergence classes.	Novel, rigorous, and distinguishable from existing dynamical systems/control theory.
Decision science	Show attractor classes predict drift, recovery, and failure thresholds.	Predictive advantage over baselines.
Economic	Apply to firms, markets, coordination failure, capital allocation, and institutional trust.	Measurable welfare, allocation, or failure-rate improvement.
AI governance	Use attractor diagnostics to reduce false state promotion in human-AI workflows.	Demonstrable safety and reliability improvements.

7. Validation Program

The added K2-K12 table improves the toy-model base but does not validate the generalized theory. Validation requires formal definitions, reproducible algorithms, datasets, baselines, and external replication.

7.1 Formal requirements

Requirement	Needed artifact
Toy-model proof	Formalize K_n cycles and basins, including leading-zero convention and excluded repdigit handling.
General definitions	Define state, rank, inversion, difference, invariant, attractor, basin, degradation, terminality, and constraint memory.
Algorithms	Reference implementation for structured signals and longitudinal state updates.
Benchmarks	Compare with Markov models, survival analysis, causal models, game theory, and expert judgment.
Falsification	Predefine where ICAD should fail or be revised.

7.2 Empirical test design

Domain	Signals	Prediction target	Evidence needed
Startups	Offer, proof, demand, distribution	Survival, revenue, conversion, pivot timing	Longitudinal data and out-of-sample prediction.
Sales/lead systems	Truth, trust, value, timing	Progress, silence, degradation, closure	Touch history and outcome labels.
Organizations	Strategy, incentives, execution, feedback	Drift, bottleneck, collapse, renewal	Operational metrics and event logs.
AI workflows	Role, context, constraint, output	False confidence, correction cycles, task success	Trace logs and verified outcomes.
Policy systems	Need, capacity, trust, implementation	Adoption, backlash, failure mode	Intervention records and response metrics.

7.3 Minimum evidence threshold

A credible external claim requires at least: formal paper, open-source reference implementation, multiple datasets, preregistered predictions, baseline comparisons, independent replication, and practical intervention results.

8. Governance and Claim Safety

The K5+ expansion introduces a risk: the table can look authoritative even though the general theory remains unvalidated. The control response is explicit source-status separation and claim labeling.

8.1 State governance imported into ICAD

A single transition is not enough to name an attractor in a real system. ICAD requires repeated observations, known constraints, and trajectory evidence. The toy models are finite and deterministic; human and economic systems are not.

8.2 Constraint memory imported into ICAD

K5+ basin ecology makes path dependence visible. In governance terms, history changes reachability: after a breach, failure, or trust loss, the state-space is not identical to the pre-breach state-space.

8.3 Claim labels

Claim type	Allowed language	Forbidden language
Toy model	K2-K12 cycles were locally computed under a stated convention.	This proves every real system has the same cycles.
Conceptual model	K9/ICAD proposes a diagnostic form.	ICAD is established science.
Nobel framing	Potentially prize-level if formalized and validated at scale.	Nobel-worthy now by assertion.
Business application	Can be used as a diagnostic lens.	Guaranteed to predict sales or markets.
Governance application	Can structure proof, state, and constraint questions.	Can replace human judgment or institutional authority.

8.4 Risk controls

Risk	Control
Over-mystification	Keep mod 9 energy-flow as metaphor; preserve arithmetic definitions.
False authority from tables	Label K2-K12 as toy-model computation only.
Unfalsifiability	Define predictions and failure cases before deployment.
Confirmation bias	Use blinded or preregistered evaluation where possible.
Constraint erasure	Keep prior states and failed predictions in the record.

9. Application Templates

The update changes application logic: do not assume one attractor. First identify whether the target system behaves like K3/K4 fixed convergence, K2/K7/K9 cycle behavior, or K5+ multi-basin ecology.

9.1 Business attractor read

Signals: Offer, Proof, Audience, Distribution

Possible attractors:

- fixed convergence: stable offer-market fit
- short cycle: repeated interest without purchase
- long cycle: complex delayed conversion path
- multi-basin: different segments converge differently
- zero collapse: no differentiation or no demand

9.2 AI workflow attractor read

Signals: Role, Context, Constraint, Output

Attractor classes: unclear role, missing context, loose constraint, wrong format, false confidence, stable execution, or repeated corre

9.3 Lead/trust attractor read

Signals: Truth, Value, Trust, Timing

If trust degrades, future motion is constrained. Interest does not erase impact; engagement does not restore permission.

9.4 Institutional collapse early-warning

Signals: Mission, Incentives, Execution, Feedback

Attractor classes: bureaucratic drift, incentive capture, execution stall, feedback blindness, ethical breach, stabilized adaptation.

9.5 Personal decision system

Signals: Desire, Fear, Discipline, Environment

Attractor classes: stable practice, resistance cycle, repetition without movement, environmental bottleneck, degradation, or identity-l

10. Roadmap, Change Control, and Coherence Record

10.1 Research roadmap

Phase	Deliverable	Exit condition
0	v0.1 base theory artifact	Core K9/ICAD/Nobel framing exists.
0.2	K2-K12 toy-model ladder	This PDF embeds the adjacent toy models and local cycle inventory.
1	Formal notation paper	Definitions and proof targets written academically.
2	Reference algorithm	Code computes attractor classes from structured input data.
3	Pilot datasets	At least two domains have labelled longitudinal data.
4	Benchmark study	Model beats or complements baselines on prediction/diagnosis.
5	External review	Independent analysts reproduce findings.
6	Economic-impact study	Measured reduction in failure, waste, misallocation, or governance error.

10.2 Change control record

Field	Value
Version	0.2.0
Status	Draft research framing
Author	TOMS, JANSONS
Effective date	2026-05-12
Supersedes	K9_Attractor_Theory_TOMS_JANSONS_v0_1.pdf
Change type	Minor expansion: new governed content without breaking the base
Summary	Adds K2/K3/K4/K5+ toy-model ladder, K2 cycle graphic, K2-K12 cycle inventory, and updated sections.
Ratification	Not ratified; requires review, checksum, and external validation before promoted status

10.3 Coherence check record

Check	Result
Core base preserved	Pass - v0.1 structure remains: base lesson, K9 form, formal model, geometry, Nobel framing, validation, governance, applications, roadmap.
New sections embedded	Pass - K2-K12 ladder and cycle inventory added.
Known/inferred/not proven separated	Pass - local toy computation separated from generalized theory.
Authorship	Pass - TOMS, JANSONS in metadata, header, footer, and change control.
Versioning	Pass - v0.2.0 marked as minor expansion superseding v0.1.
Overclaim prevention	Pass - no Nobel award, peer review, or empirical validation claimed.
Layout verification	Pass after render-based QA; any defects require revision before delivery.

10.4 Final compressed doctrine

K2 teaches cycle. K3 teaches minimal fixed convergence. K4 teaches canonical fixed convergence. K5+ teaches basin ecology. ICAD asks which of these structures a real bounded system resembles under pressure, then governs action from evidence rather than assertion.

Appendix A. K2-K12 Kaprekar Cycle Inventory

Convention: base-10, fixed digit length n, leading zeros retained, sorted digit-multiset state graph. The trivial zero cycle is listed separately or omitted from the non-trivial count. These are local computation results generated inside the artifact workflow.

A.2. K2 cycle inventory

Type	Length	Cycle
Trivial zero	1	00
Non-trivial	5	09 -> 81 -> 63 -> 27 -> 45

A.3. K3 cycle inventory

Type	Length	Cycle
Trivial zero	1	000
Non-trivial	1	495

A.4. K4 cycle inventory

Type	Length	Cycle
Trivial zero	1	0000
Non-trivial	1	6174

A.5. K5 cycle inventory

Type	Length	Cycle
Trivial zero	1	00000
Non-trivial	2	53955 -> 59994
Non-trivial	4	61974 -> 82962 -> 75933 -> 63954
Non-trivial	4	62964 -> 71973 -> 83952 -> 74943

A.6. K6 cycle inventory

Type	Length	Cycle
Trivial zero	1	000000
Non-trivial	1	549945
Non-trivial	1	631764
Non-trivial	7	420876 -> 851742 -> 750843 -> 840852 -> 860832 -> 862632 -> 642654

A.7. K7 cycle inventory

Type	Length	Cycle
Trivial zero	1	0000000
Non-trivial	8	7509843 -> 9529641 -> 8719722 -> 8649432 -> 7519743 -> 8429652 -> 7619733 -> 8439552

A.8. K8 cycle inventory

Type	Length	Cycle
Trivial zero	1	00000000
Non-trivial	1	63317664
Non-trivial	1	97508421
Non-trivial	3	64308654 -> 83208762 -> 86526432
Non-trivial	7	43208766 -> 85317642 -> 75308643 -> 84308652 -> 86308632 -> 86326632 -> 64326654

A.9. K9 cycle inventory

Type	Length	Cycle
Trivial zero	1	000000000
Non-trivial	1	554999445
Non-trivial	1	864197532
Non-trivial	14	753098643 -> 954197541 -> 883098612 -> 976494321 -> 874197522 -> 865296432 -> 763197633 -> 844296552 -> 762098733 -> 964395531 -> 863098632 -> 965296431 -> 873197622 -> 865395432

A.10. K10 cycle inventory

Type	Length	Cycle
Trivial zero	1	0000000000
Non-trivial	1	6333176664
Non-trivial	1	9753086421
Non-trivial	1	9975084201
Non-trivial	3	6431088654 -> 8732087622 -> 8655264432
Non-trivial	3	6433086654 -> 8332087662 -> 8653266432
Non-trivial	3	6543086544 -> 8321088762 -> 8765264322
Non-trivial	3	9751088421 -> 9775084221 -> 9755084421
Non-trivial	7	4332087666 -> 8533176642 -> 7533086643 -> 8433086652 -> 8633086632 -> 8633266632 -> 6433266654

A.11. K11 cycle inventory

Type	Length	Cycle
Trivial zero	1	00000000000
Non-trivial	1	86431976532
Non-trivial	5	86420987532 -> 96641975331 -> 88431976512 -> 87641975322 -> 86541975432
Non-trivial	8	76320987633 -> 96442965531 -> 87320987622 -> 96653954331 -> 86330986632 -> 96532966431 -> 87331976622 -> 86542965432

A.12. K12 cycle inventory

Type	Length	Cycle
Trivial zero	1	000000000000
Non-trivial	1	555499994445
Non-trivial	1	633331766664
Non-trivial	1	975330866421
Non-trivial	1	997530864201
Non-trivial	1	999750842001
Non-trivial	3	643110888654 -> 877320876222 -> 865552644432
Non-trivial	3	643310886654 -> 873320876622 -> 865532664432
Non-trivial	3	643330866654 -> 833320876662 -> 865332666432
Non-trivial	3	654310886544 -> 873210887622 -> 876552644322
Non-trivial	3	654330866544 -> 833210887662 -> 876532664322
Non-trivial	3	655430865444 -> 832110888762 -> 877652643222
Non-trivial	3	975110888421 -> 977750842221 -> 975550844421
Non-trivial	3	975310886421 -> 977530864221 -> 975530864421
Non-trivial	3	975510884421 -> 977510884221 -> 977550844221
Non-trivial	3	997510884201 -> 997750842201 -> 997550844201
Non-trivial	7	433320876666 -> 853331766642 -> 753330866643 -> 843330866652 -> 863330866632 -> 863332666632 -> 643332666654

Appendix B. Local Computation Receipt

The following pseudocode documents the local verification method used to generate the K2-K12 table.

```
for n in 2..12:
  states = all sorted digit multisets of length n
  for each state s:
    desc = s
    asc = reverse(s)
    output = zero_pad(int(desc) - int(asc), n)
    next_state = sort_desc(output)
  detect directed cycles in the finite graph
  record fixed points and cycles
```

This receipt supports the toy-model table only. It does not validate ICAD in economics, governance, AI, or human decision systems.